

PATH-SPACE COTANGENT RIGIDITY, UNIVERSAL CONTRACTIONS, AND TANGENT REPRESENTATIONS

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ABSTRACT. Physical platforms form a labelled coproduct and their source spaces a locally convex direct sum. Explicit rate–moment coercivity makes weighted strict pressures finite and their phase sets compact. Every mechanical observable is a continuous or exponentially approximable typed contraction. Finite likelihoods are exact conditional-expectation martingales on complete history filtrations and converge through the A4 conditional-kernel theorem to diffusion optional projections, Girsanov densities, and BSDEs. Linearized pressure and compressed memory are paired in one declared Doob Hilbert space with one orthogonal projection.

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1. ROUND-SIX POSITIVE CLOSURE: COERCIVE TYPED CONTRACTIONS AND HISTORY–DOOB REPRESENTATIONS

Every physical platform retains its label. Weighted pressures are considered only on source domains controlled by an explicit rate–moment coercivity

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inequality. Finite likelihoods are exact conditional-expectation martingales on the complete history filtration. The memory identity is paired in one declared Doob Hilbert space and is constructed directly from the Doob generator; no unproved identification with an untitled Koopman compression is used.

1.1. The platform coproduct and source direct sum. Let $\mathfrak{A}$ index the constructed platforms: the A1 Hamiltonian port system, the Sinai collision/physical path system, and the hard-sphere Boltzmann–Grad density/contact system. For each α , let Ω_α be its Polish graph-completed path space, let $W_\alpha \geq 1$ be continuous with compact sublevels, and let I_α be its good rate. Assume the proved platform coercivity

$$I_\alpha(\nu) \geq a_\alpha \nu(W_\alpha) - b_\alpha, \quad a_\alpha > 0.$$

For hard spheres this is read on each bounded Maxwellian source chart and then by lower-semicontinuous exhaustion; for Sinai paths it follows from the A3 recurrent–defect rate.

Define

$$\mathfrak{P} = \bigsqcup_{\alpha \in \mathfrak{A}} \{\alpha\} \times \mathcal{P}_{W_\alpha}(\Omega_\alpha), \quad \mathfrak{J}(\alpha, \nu) = I_\alpha(\nu).$$

The source space is the locally convex direct sum

$$\mathfrak{X} = \bigoplus_{\alpha \in \mathfrak{A}} (C_{W_\alpha, 0}(\Omega_\alpha), \beta_{W_\alpha}),$$

whose continuous dual is the product of the weighted Radon measure spaces. A physical phase occupies one labelled component; the product dual is a bookkeeping object and does not randomize the platform label.

Theorem 1.1 (Coercive strict pressure and phase compactness). *For $F \in C_{W_\alpha, 0}$ with $\|F/W_\alpha\|_\infty < a_\alpha$,*

$$Q_\alpha(F) = \sup_{\nu} \{\nu(F) - I_\alpha(\nu)\}$$

is finite, strictly continuous on bounded subsets of every smaller source ball, and attains its supremum on a compact set. Its strict subdifferential is exactly

$$\partial Q_\alpha(F) = \operatorname{argmax}_{\nu} \{\nu(F) - I_\alpha(\nu)\},$$

a compact convex set of countably additive weighted Radon probabilities. The componentwise pressures define a continuous convex functional on each bounded set of the locally convex direct sum $\mathfrak{X}$.

Proof. Choose $c < a_\alpha$ with $|F| \leq cW_\alpha + C$. Then

$$\nu(F) - I_\alpha(\nu) \leq C + b_\alpha - (a_\alpha - c)\nu(W_\alpha),$$

so maximizing sequences have uniformly bounded W_α -moment and lie in one compact rate sublevel. Strict convergence on a bounded source set is uniform on that compact set, proving continuity and attainment. The standard Fenchel equality shows that every maximizer is a subgradient. Conversely,

the weighted-strict dual is the Radon measure space; testing constants and positive functions makes a subgradient a probability, and Fenchel equality makes it a maximizer. Direct-sum continuity is componentwise because bounded sets have only finitely many active components. $\square$

1.2. Typed contraction maps. A platform map $C_\alpha : \mathcal{P}_{W_\alpha}(\Omega_\alpha) \rightarrow E_\alpha$ is admitted if it is continuous on rate sublevels or has continuous approximants $C_{\alpha,n}$ satisfying

$$\lim_{n \rightarrow \infty} \limsup_{\varepsilon \rightarrow 0} \frac{1}{\mu_\varepsilon} \log \mathbb{P}_\varepsilon \{d(C_\alpha, C_{\alpha,n}) > \delta\} = -\infty.$$

For Sinai paths, A3 supplies such shields for collision expansion, physical clock, occupation, homology, and bounded local work. For hard spheres, density/contact linear maps are continuous in the B2 weighted weak topology. The A1 current is continuous on its common symbolic factor and has the physical trace realization proved there.

Theorem 1.2 (Platform-labelled contraction functor). *Every admitted map has rate*

$$I_{\alpha,C}(y) = \inf_{C_\alpha(\nu)=y} I_\alpha(\nu).$$

For a continuous linear observable map with adjoint C_α^* ,

$$Q_{\alpha,C}(\eta) = Q_\alpha(C_\alpha^* \eta).$$

First and second pressure derivatives commute with the map on every local analytic phase chart. A cross-platform statement is valid only after a separate scaling theorem supplies an actual map between the two labelled microscopic law families.

Proof. Apply the ordinary or extended contraction theorem on one open-and-closed labelled component. The finite-volume pressure identity follows from $\langle \eta, CX \rangle = \langle C^* \eta, X \rangle$; local uniform analytic convergence permits differentiation. Since the coproduct topology prevents a sequence from changing labels, no cross-platform contraction exists without a separate microscopic scaling map. $\square$

1.3. Closed coboundary cotangents and pressure responses. On a stationary path component let $\mathcal{A}_W$ be the weighted dynamically Hölder algebra and let

$$\mathcal{N}_W = \overline{\text{span}}^{\mathcal{A}_W} (\mathbb{R}\mathbf{1} + \{U - U \circ \Theta_t : U \in \mathcal{A}_W, t \in \mathbb{Q}\}).$$

Theorem 1.3 (Path cotangent quotient). *The quotient $\mathcal{A}_W/\mathcal{N}_W$ is Hausdorff. Two potentials have the same integral under every invariant finite-weight phase if and only if they have the same quotient class. On a simple Sinai phase, the first pressure derivative is the prepared path law and the second is the absolutely convergent Green–Kubo covariance derived from the A2 spectral projector. On a regular hard-sphere phase, the corresponding*

derivatives are the B3 exact mean and quotient covariance. At coexistence, the strict subdifferential is the compact maximizing phase set of Theorem 1.1.

Proof. Closedness makes the quotient Hausdorff. Invariant laws annihilate rational-time coboundaries. Conversely, if a quotient class is nonzero, Hahn–Banach supplies a continuous strict-dual separator. Averaging it over rational time shifts yields an invariant signed Radon functional; its Jordan components, after balancing their equal total masses, give invariant weighted probabilities separating the class. The response formulas follow by differentiating the A2 simple eigenvalue or the B2/B3 local analytic pressure. The strict subgradient statement is Theorem 1.1. $\square$

1.4. Exact history likelihood martingales. Let X^ε be a resolved slow functional of a deterministic platform and let $\mathcal{F}_t^{\text{hist}}$ be its complete resolved-history filtration. For a bounded Lipschitz functional Φ of a finite future enhanced path window, define

$$M_t^\varepsilon = \frac{\mathbb{E}[e^{\theta\Phi(X^\varepsilon)} \mid \mathcal{F}_t^{\text{hist}}]}{\mathbb{E}e^{\theta\Phi(X^\varepsilon)}}.$$

Lemma 1.4 (Exact finite likelihood). *The process M^ε is a positive mean-one uniformly integrable martingale. It is represented by the complete-history kernel*

$$M_t^\varepsilon = \frac{K_{t,T}^\varepsilon(e^{\theta\Phi})(H_t^\varepsilon)}{\mathbb{E}e^{\theta\Phi(X^\varepsilon)}}.$$

No present-state Markov identity is assumed at finite ε .

Proof. The numerator is the optional projection of the bounded positive terminal variable. The conditional-expectation tower gives the martingale property, and boundedness of Φ gives uniform integrability. Regular conditional probability on the standard Borel history space gives the kernel representation. $\square$

The A4 conditional-kernel theorem states that on compact history sets these kernels converge uniformly, stably with initial history variables, to the kernel of the area-corrected rough diffusion. B2/B3 exponential containment supplies compact-history localization on the hard-sphere side whenever a separate kinetic-to-diffusion scaling is used.

Theorem 1.5 (Optional-projection convergence). *For every bounded Lipschitz finite-window functional Φ ,*

$$(X^\varepsilon, M^\varepsilon) \implies (X, M)$$

stably in path space, where

$$M_t = \frac{\mathbb{E}[e^{\theta\Phi(X)} \mid \mathcal{F}_t^X]}{\mathbb{E}e^{\theta\Phi(X)}}.$$

If $\Phi = \phi(X_T)$, the limit depends on the present diffusion state; for a general future-path functional it is the corresponding path-kernel on the augmented Markov state and is not replaced by an endpoint semigroup.

Proof. Localize to a compact history set whose complement has arbitrarily small probability uniformly in ε . On that set, A4 gives uniform conditional-kernel convergence for the enhanced slow path. The denominator converges by bounded convergence. Stable convergence follows by multiplying with any bounded initial-history variable, which is included in the A4 conditional theorem. Remove the localization. The final distinction is simply whether the limiting conditional functional is measurable from the present Markov state or requires a path-augmented state. $\square$

1.5. Girsanov and BSDE on a smooth diffusion chart. Assume now

$$dX_t = b(X_t)dt + \sigma(X_t)dW_t, \quad A = \sigma\sigma^T,$$

with bounded C_b^3 coefficients, uniformly elliptic A , and $\phi \in C_b^3$. Put

$$u(t, x) = \theta^{-1} \log \mathbb{E}_{t,x} e^{\theta\phi(X_T)}.$$

Theorem 1.6 (Likelihood–Girsanov–BSDE representation). *The likelihood limit in Theorem 1.5 is*

$$M_t = \mathcal{E} \left(\int_0^t \theta \sigma^T Du(s, X_s) dW_s \right).$$

Under the tilted law,

$$dX_t = (b + \theta ADu)(t, X_t)dt + \sigma(X_t)dW_t^\theta.$$

Moreover $(Y_t, Z_t) = (u(t, X_t), \sigma^T Du(t, X_t))$ is the unique bounded solution of

$$Y_t = \phi(X_T) + \int_t^T \frac{\theta}{2} |Z_s|^2 ds - \int_t^T Z_s dW_s.$$

For a rough diffusion inherited from A4, its area anomaly is included in the effective drift b .

Proof. Uniform ellipticity and the smooth bounded coefficients make u the bounded classical solution of

$$\partial_t u + b \cdot Du + \frac{1}{2} A : D^2 u + \frac{\theta}{2} Du^T A Du = 0.$$

Itô's formula identifies the normalized terminal exponential with the stated stochastic exponential. Girsanov's theorem gives the drift shift. Applying Itô's formula to $u(t, X_t)$ gives the BSDE, and the exponential transform gives uniqueness in the bounded class. The deterministic rough-area correction contributes its standard Lie-bracket drift before this diffusion calculus is applied. $\square$

1.6. One Hilbert-space Doob memory identity. Fix one exposed potential Ψ . Let $\tilde{P}_t^\Psi$ be the genuine A4 eigenfunction Doob semigroup and let ν_Ψ^D be its invariant phase. Work solely in

$$\mathcal{H}_D = L^2(\nu_\Psi^D),$$

with generator L_D . Let $V \subset D(L_D^2)$ be finite dimensional and let P_D be its $\mathcal{H}_D$ -orthogonal projection. Define

$$\mathcal{E}_t^{\Psi, \theta} F = \theta^{-1} \log \tilde{P}_t^\Psi(e^{\theta F}).$$

Theorem 1.7 (Typed pressure-tangent/memory commutation). *For $A, B \in V$,*

$$\int_0^\infty e^{-zt} \left\langle B, D\mathcal{E}_t^{\Psi, \theta}(0)A \right\rangle_{\mathcal{H}_D} dt = \left\langle B, P_D(z - L_D)^{-1} P_D A \right\rangle_{\mathcal{H}_D}.$$

The Doob compressed-memory transform is therefore

$$\widehat{K}_D(z) = zI - P_D L_D P_D - [P_D(z - L_D)^{-1} P_D]^{-1}.$$

It is the same operator obtained by first linearizing the nonlinear Doob pressure in $\mathcal{H}_D$ and then applying the A_4 compressed-correlation Volterra construction. Source differentiation on either side gives the same Duhamel insertion.

Proof. Direct differentiation at zero gives

$$D\mathcal{E}_t^{\Psi, \theta}(0)A = \tilde{P}_t^\Psi A,$$

not a statewise normalized Feynman–Kac ratio. Pairing in the declared Hilbert space and taking the Laplace transform gives the compressed resolvent. A_4 's Volterra theorem applies directly to the strongly continuous Doob semigroup and the orthogonal projection P_D , yielding the memory formula. Both source derivatives are obtained from

$$\partial_\eta e^{t(L_D + \eta V)}|_{\eta=0} = \int_0^t e^{(t-s)L_D} V e^{sL_D} ds,$$

so the diagram commutes. $\square$

A Feynman–Kac generator before the Doob transform may be related by multiplication with the bounded positive eigenfunction, but no equality with an untitled Koopman memory is asserted unless a bounded unitary intertwiner and the transformed orthogonal projection have been explicitly supplied.

1.7. Conditional entropic calibration. Cash additivity and strong time consistency classify a smooth utility-generated certainty equivalent as affine or entropic with one constant coefficient. Equality with a mechanical preparation field follows only when the two normalized likelihood cocycles are required to coincide in common work units.

Theorem 1.8 (Round-six C2 closure). *The theory has coercive platform-labelled pressures, countably additive strict phase duals, closed path cotangent quotients, and typed contraction maps. Finite likelihoods are exact complete-history martingales and converge through the A_4 conditional-kernel theorem to the appropriate diffusion optional projections, yielding the smooth Girsanov and BSDE representations. The pressure-tangent/memory identity is constructed in one Doob Hilbert space with one orthogonal projection.*

Proof. The strict and contraction assertions are Theorems 1.1, 1.2, and 1.3. Likelihood convergence and its stochastic representations are Theorems 1.5 and 1.6. The memory statement is Theorem 1.7. The calibration qualification was proved in the final subsection. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-three dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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